

Open Source Solutions for Building Specialised Language Models: An Overview

Specialised language models score over large language models in various ways. What's more, there are a range of open source solutions you can choose from to build a reliable model.



A large language model (LLM) has millions of parameters whereas a small language model has significantly fewer parameters, uses less resources and is optimised for a specific domain. The specialised language model (SLM) can be small or large in model size but focuses on specific fields like law, healthcare, and so on.

Creating a specialised language model using multiple LLM sources

The process of developing an SLM involves harnessing the strengths of multiple LLMs to filter data effectively. This requires several steps, which are outlined below.

Data collection: The first step is to gather a diverse set of data from various sources, including domain-specific databases, scientific journals, articles, and generic data

repositories. The goal is to assemble a comprehensive dataset that encompasses both specialised and general knowledge.

Data preprocessing: Data preprocessing is essential for cleaning and organising the collected data. This step involves removing duplicates, irrelevant information, and noise. Techniques such as tokenization, stemming, and lemmatization are employed to standardise the text.

Data filtering: To create an effective SLM, it is crucial to filter out domain-specific data from generic information. This can be achieved by leveraging multiple LLMs, each trained on different datasets. These models can be used to classify and segregate data based on their relevance and context.

Model training: Once the data is filtered, the next step is to train the SLM. This involves fine-tuning the selected LLMs on the domain-specific dataset. Techniques such as